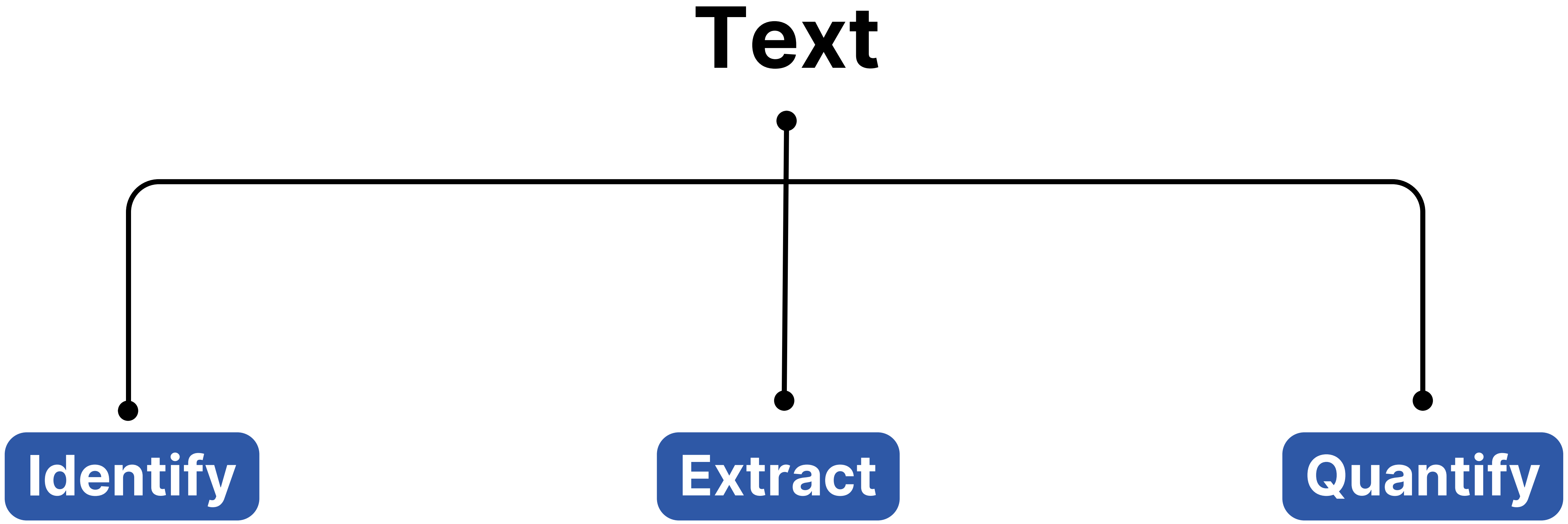
University of the Philippines | 2026

Integrating Social Context Into EmoBertGCN For Sentiment Analysis Of Filipino Political Discourse

Presented by Gabriel Virtucio

Sentiment Analysis



BertGCN Architecture

BERT Embedding initialized Document nodes

Global corpus-level relationships

Joint Learning between GCN and BERT

(Lin et al. 2021)

This model treat documents in isolation, ignoring the conversational context when applied to social media posts.

Core Idea

By adding conversation edges, we create pathways for emotional contagion to flow, giving the model more context to understand sentiment.

Theoretical Foundation

Emotional Contagion

Emotions can spread through social networks (Coviello et al., 2014). A reply to an emotional post is likely to be influenced by it.

Theoretical Foundation

Homophily

"Birds of a feather flock together." Users with similar views tend to interact, creating homophilous networks (McPherson et al., 2001).

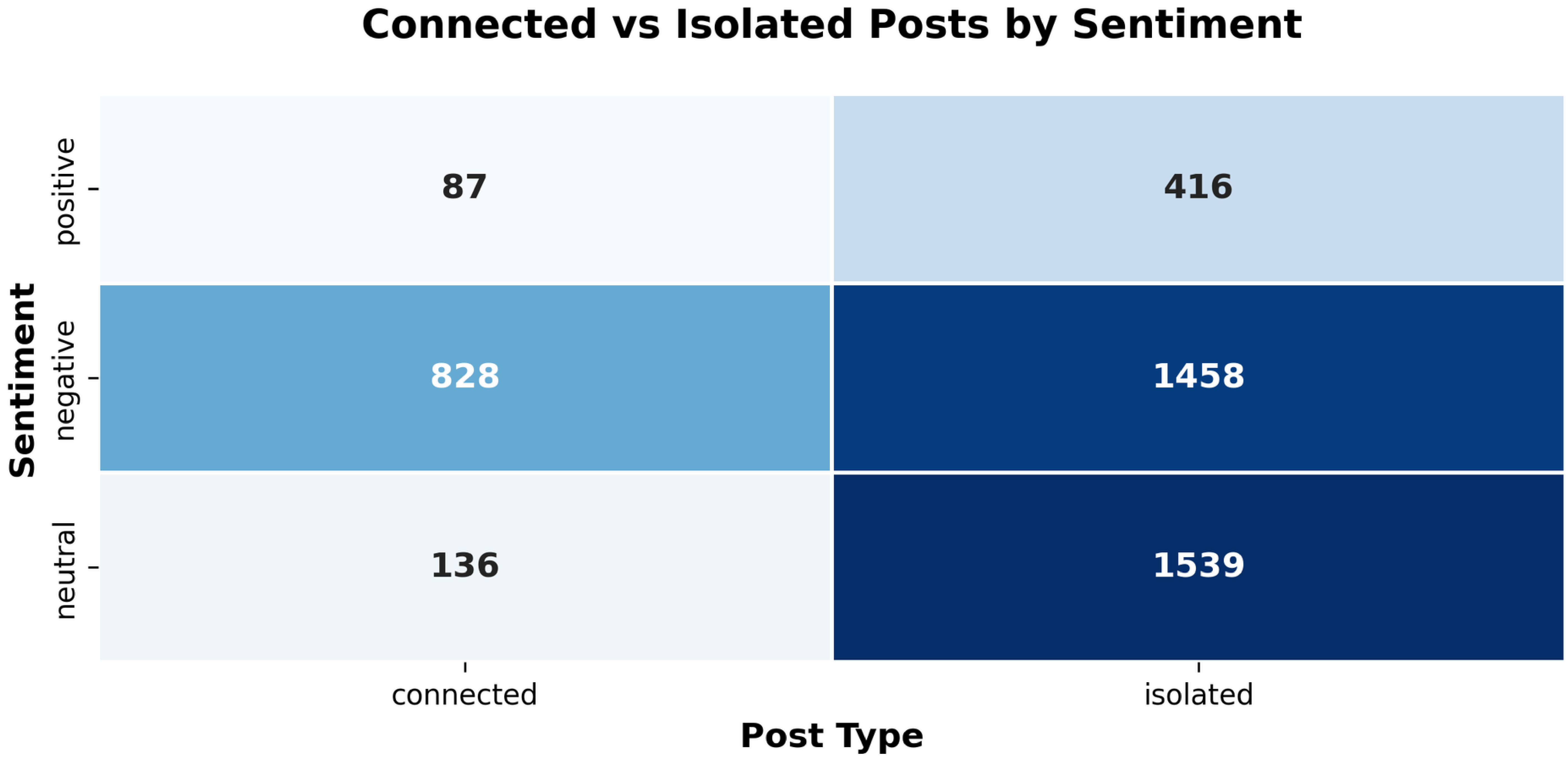
Methodology

Dataset

- Data Source: ~74k filtered tweets from the 2025 Philippine Midterm Elections.
- Sentiment Labeling: Manually annotated 4464 tweets into Positive, Negative, and Neutral.
- Inter-Annotator Agreement = 0.77

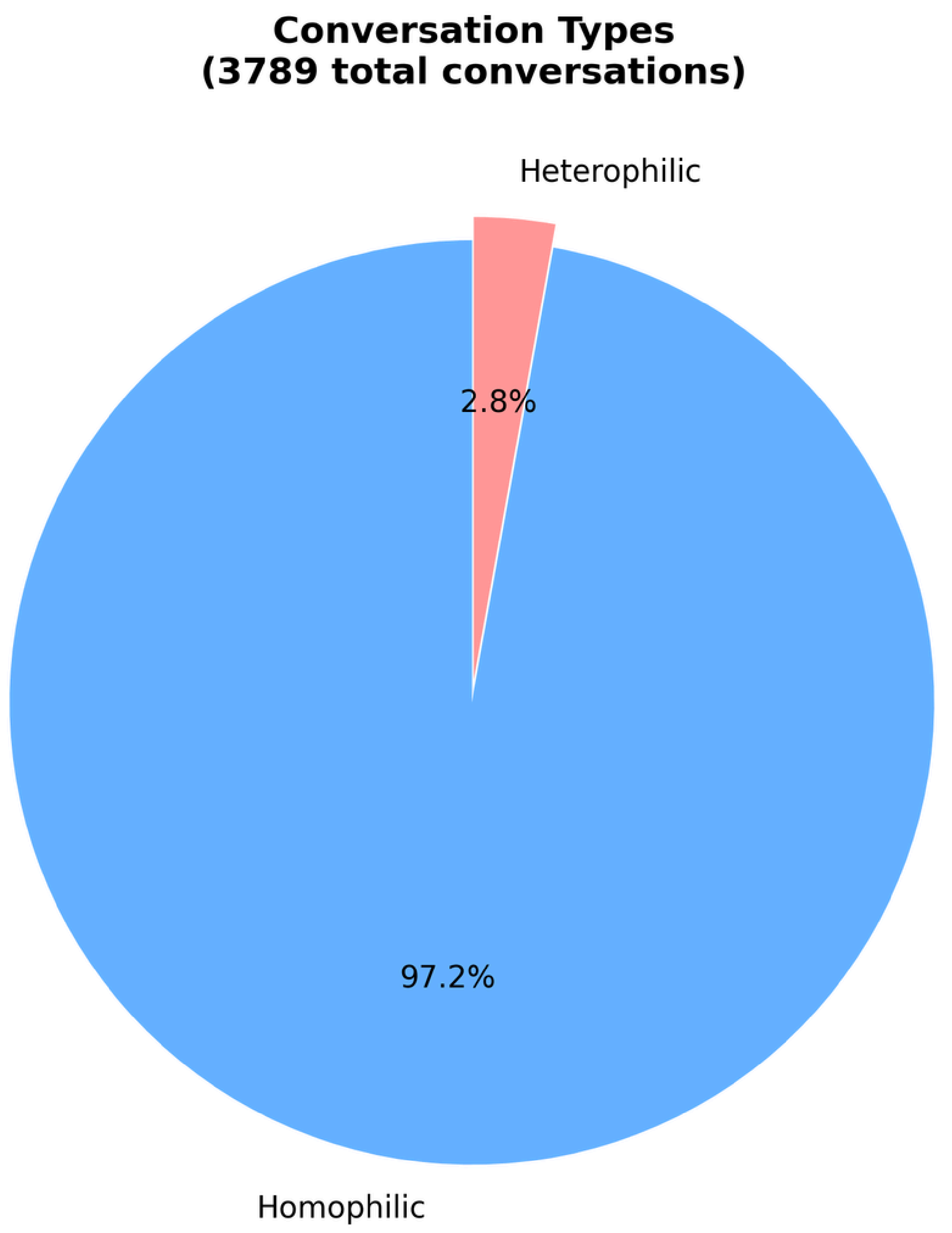
Methodology

Dataset Characteristics



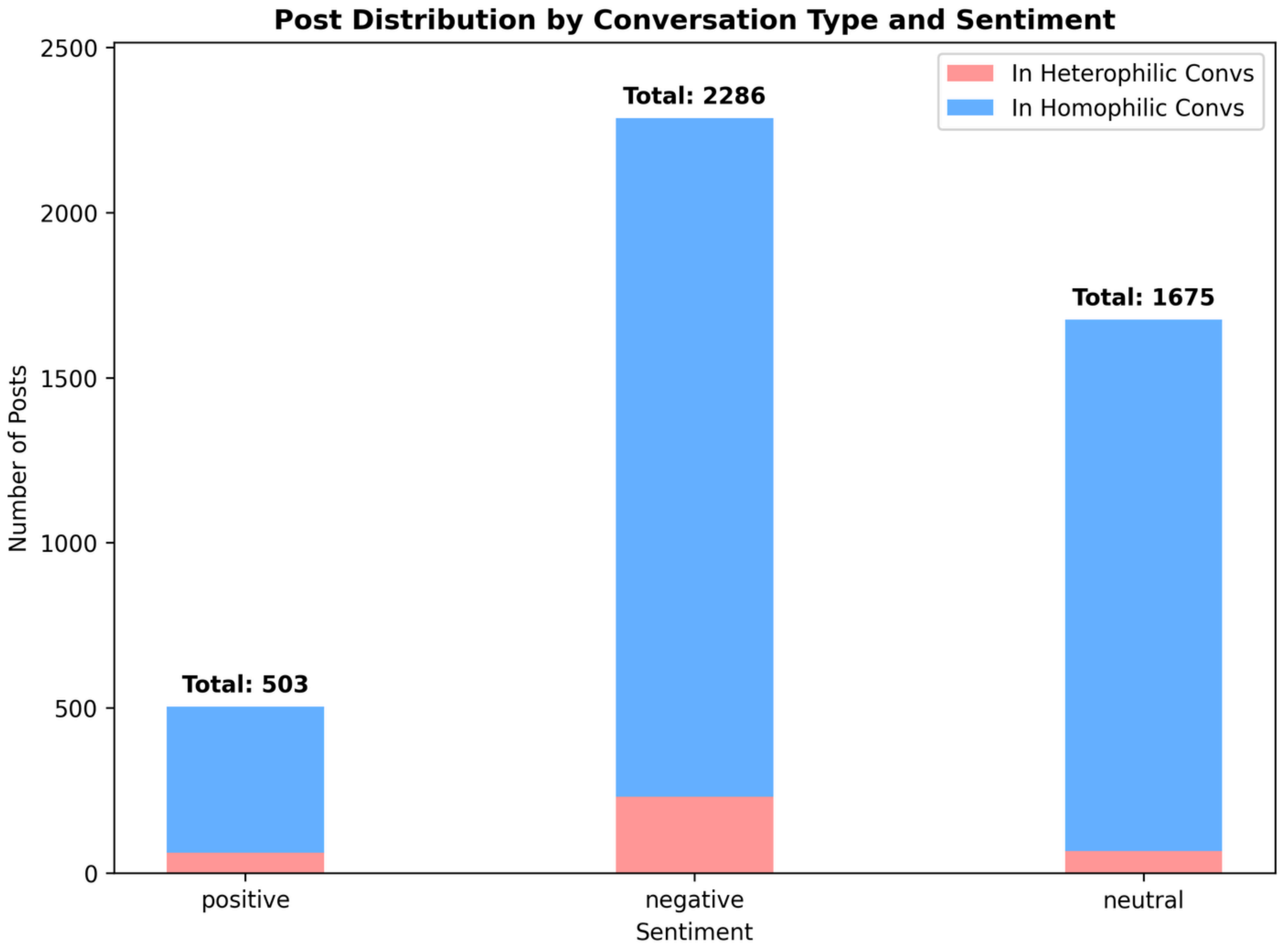
Methodology

Dataset Characteristics



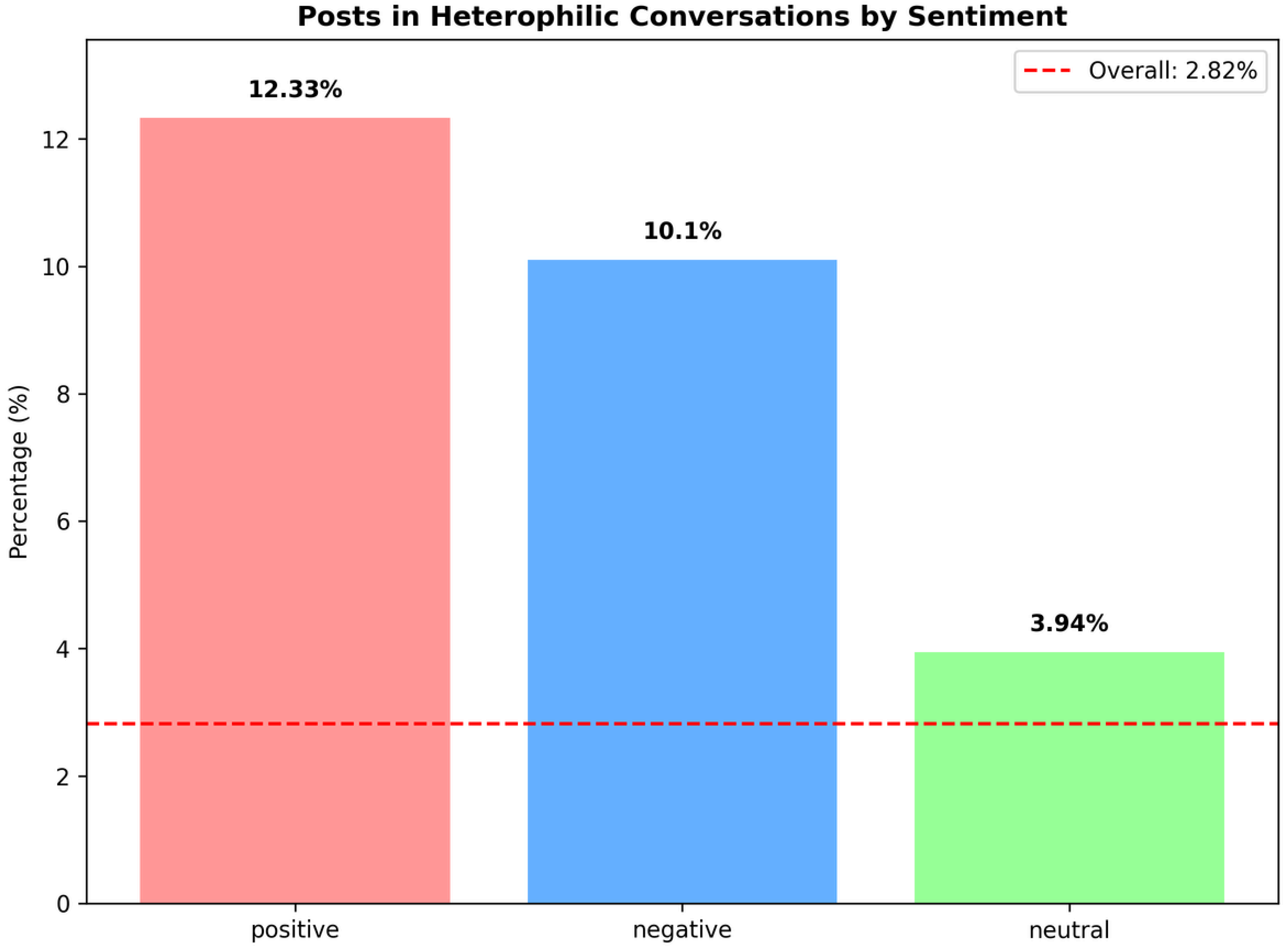
Methodology

Dataset Characteristics



Methodology

Dataset Characteristics



Methodology

Training

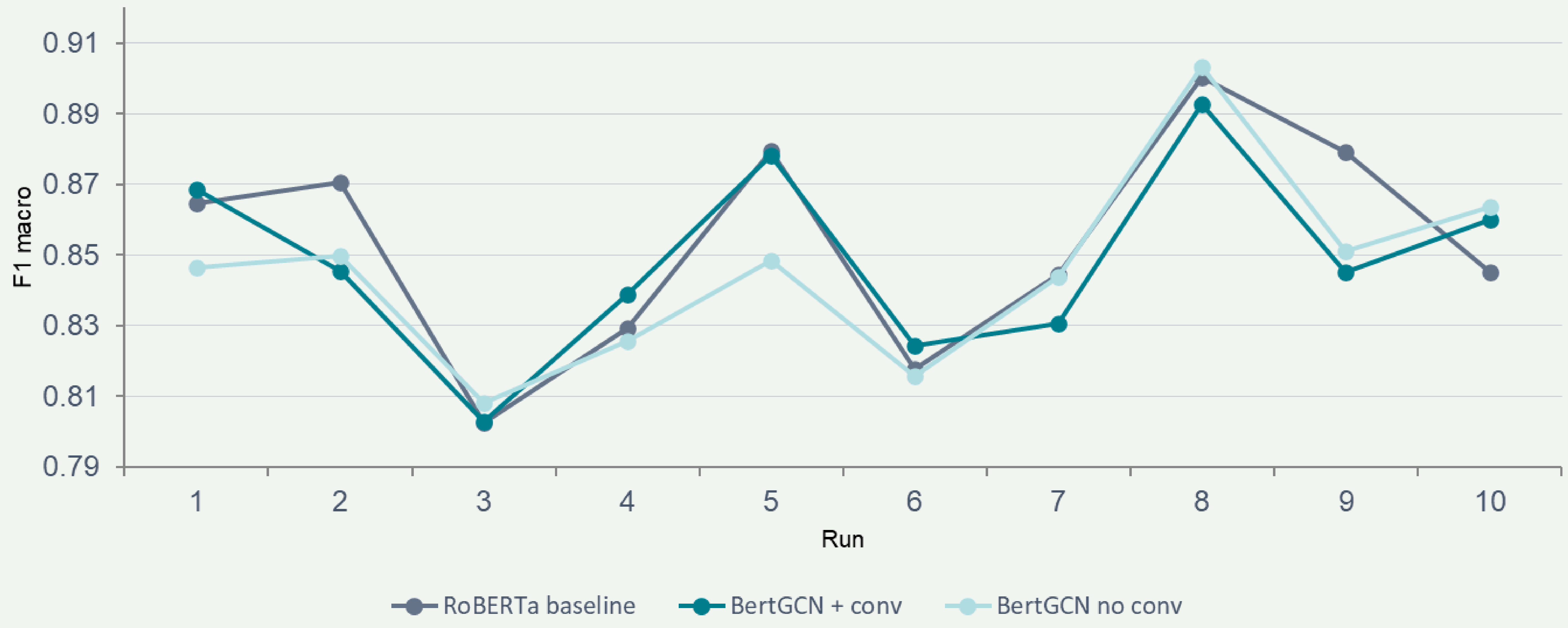
- Preprocessing: Retained emojis, hashtags, and mentions as special tokens.
- Training & Evaluation: 10-Fold Cross-Validation with Macro F1-Score as the primary metric.

Methodology

Topic Modeling

- Using BERT-LDA Hybrid Model, we discover the main topics or themes within each sentiment category (Positive, Negative, Neutral).
- LDA (Latent Dirichlet Allocation): A traditional algorithm that discovers latent topics in documents.
- Combination: Uses BERT's understanding to create better, more distinct topic clusters than LDA alone.

Results



High run-to-run variance relative to mean differences — overlapping distributions explain the null result

Results

NULL RESULT

Conversation edges does not improve BertGCN F1 performance

BertGCN+conv (0.8487) vs BertGCN no-conv (0.8456): $\Delta = +0.0031$, $p = 0.401$, Cohen's $d = 0.11$. The effect is negligible and not distinguishable from sampling noise.

NULL RESULT

BertGCN with conversation edges does not outperform the RoBERTa

BertGCN+conv (0.8487) vs RoBERTa (0.8534): $\Delta = -0.0047$, $p = 0.642$, Cohen's $d = -0.16$. The GCN component adds no measurable gain over the fine-tuned baseline.

NULL RESULT

Joint FWER result indicates the proposed model does not outperform either baseline.

Joint FWER p-value = 0.647

Results

3 clusters

UMass: -2.985362
UCI: -2.472993

4 clusters

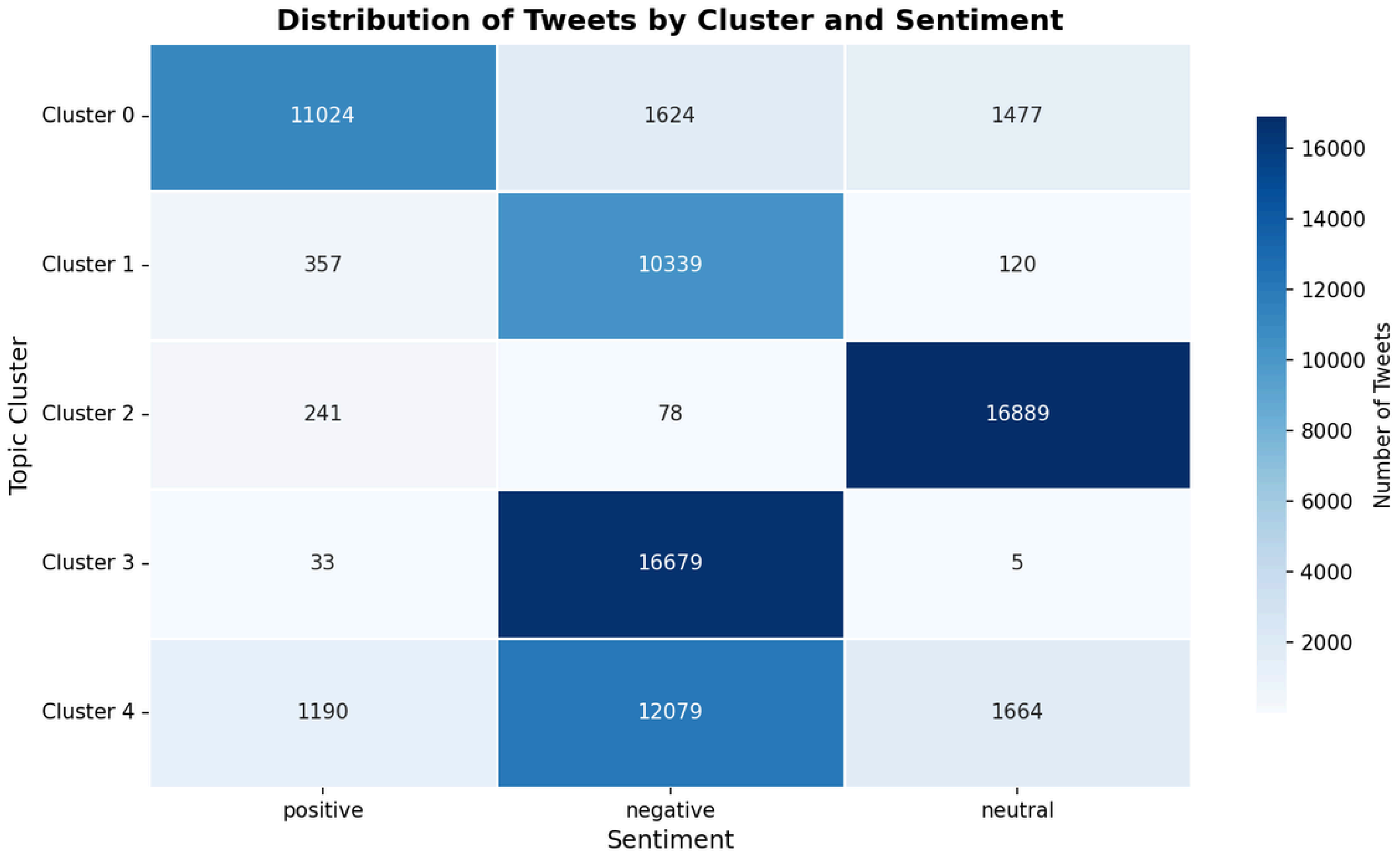
UMass: -3.150664
UCI: -2.485665

5 clusters

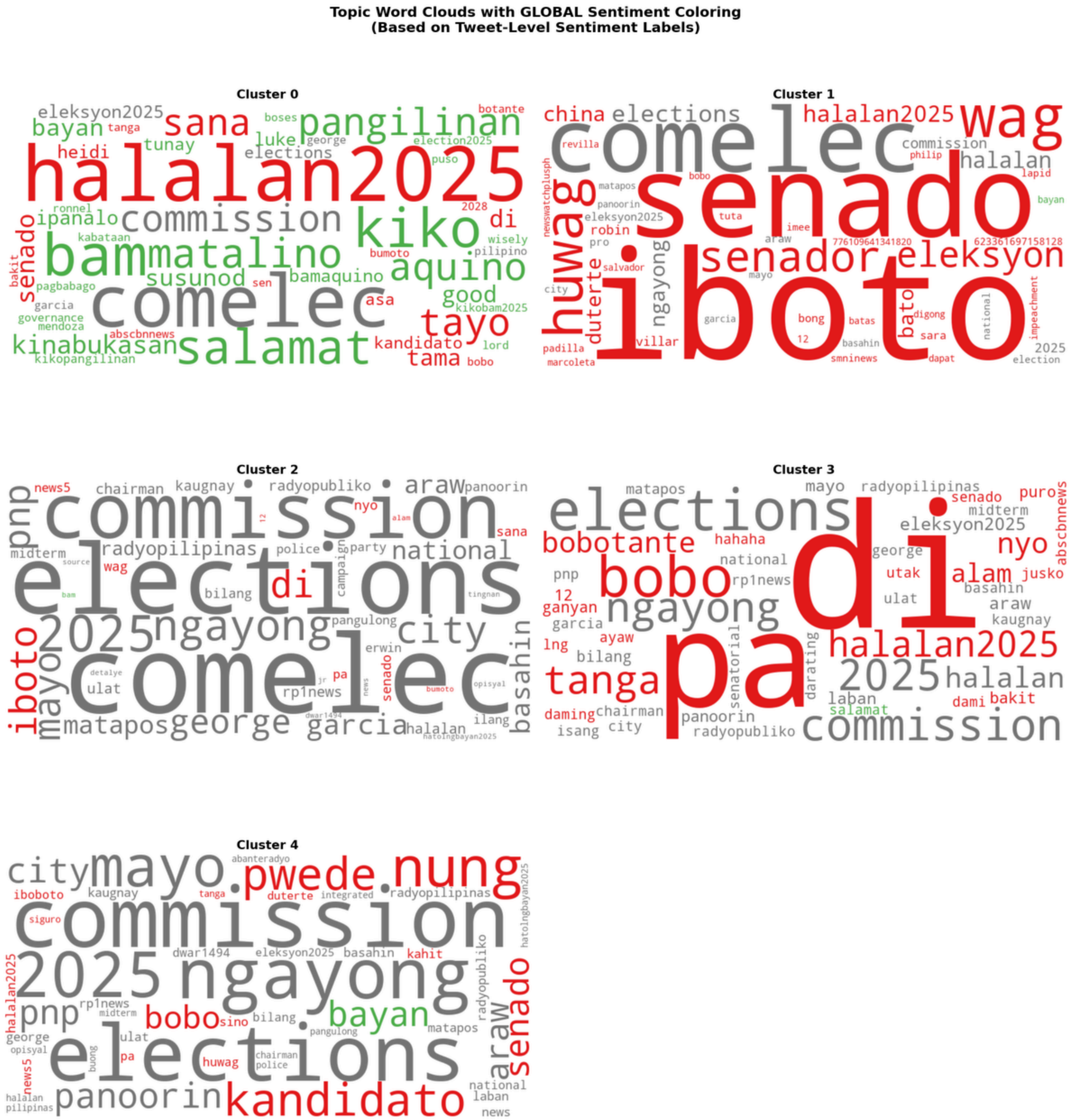
UMass: -3.35179
UCI: -1.912586

UCI measures how often pairs of top words appear together within a sliding window (10 consecutive words) while UMass measures how often top words in a topic co-occur

Results



Distribution of Tweets by Cluster and Sentiment
Topic Cluster
Sentiment
positive
negative
neutral
Number of Tweets
2000
4000
6000
8000
10000
12000
14000
16000
Cluster 0
11024
1624
1477
Cluster 1
357
10339
120
Cluster 2
241
78
16889
Cluster 3
33
16679
5
Cluster 4
1190
12079
1664



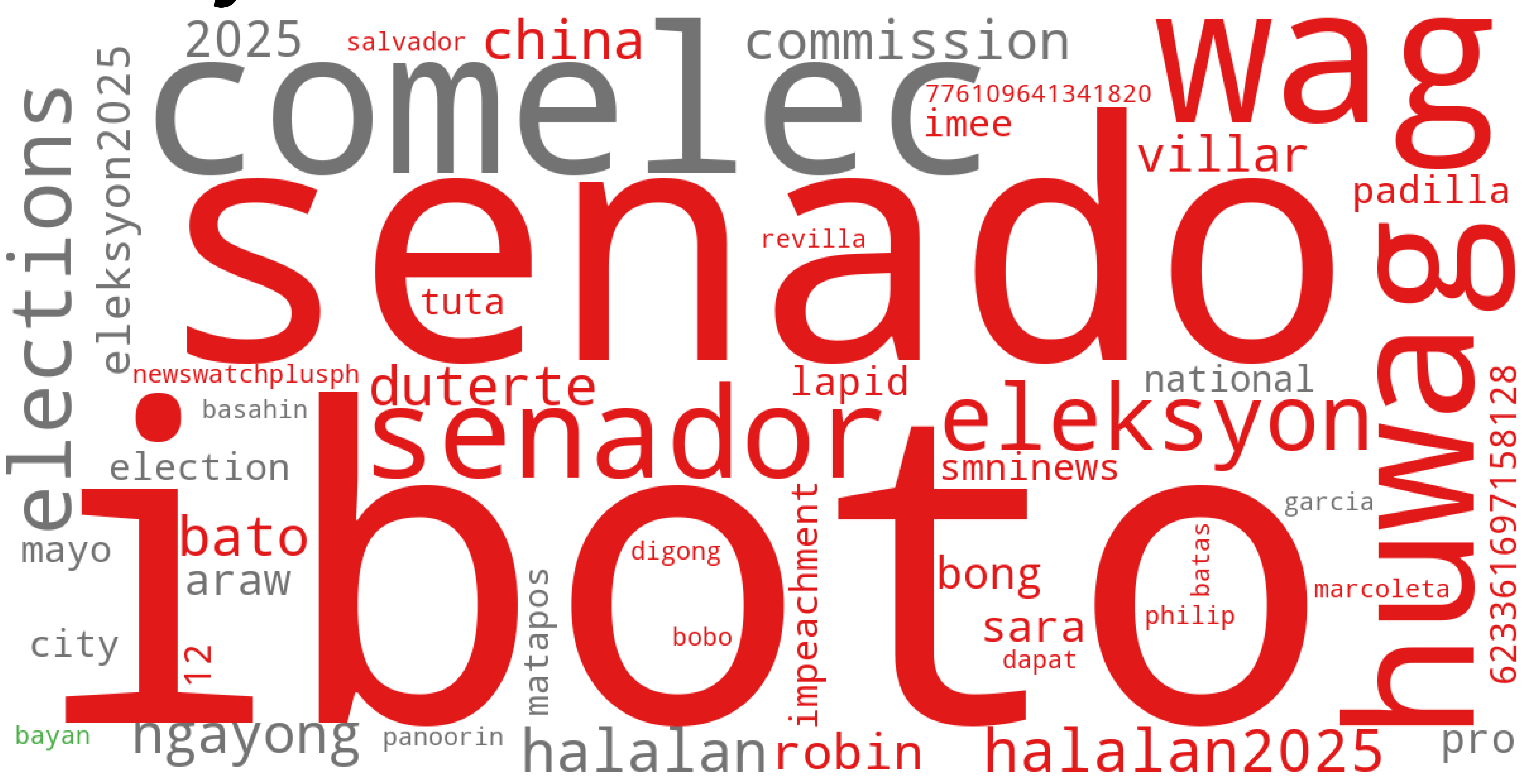
Topic Word Clouds with GLOBAL Sentiment Coloring
(Based on Tweet-Level Sentiment Labels)
Cluster 0
halalan2025 pangilinan bam matalino kiko aquino comelec salamat commission kinabukasan sana tayo bayan eleksyon2025 susunod good tama kandidato
Cluster 1
comelec senado iboto senador eleksyon halalan2025 wag huwag elections china halalan duterte ngayong
Cluster 2
commission elections comelec 2025 ngayong city iboto national radyopilipinas mayo di garcia george matapos
Cluster 3
di pa elections bobo bobotante ngayong halalan2025 2025 halalan tanga commission alam nyo
Cluster 4
commission ngayong 2025 elections mayo city pwede nung bobo bayan kandidato panoorin senado pnp

Results

Bam-Kiko Reformist Bloc (Pro-Opposition)

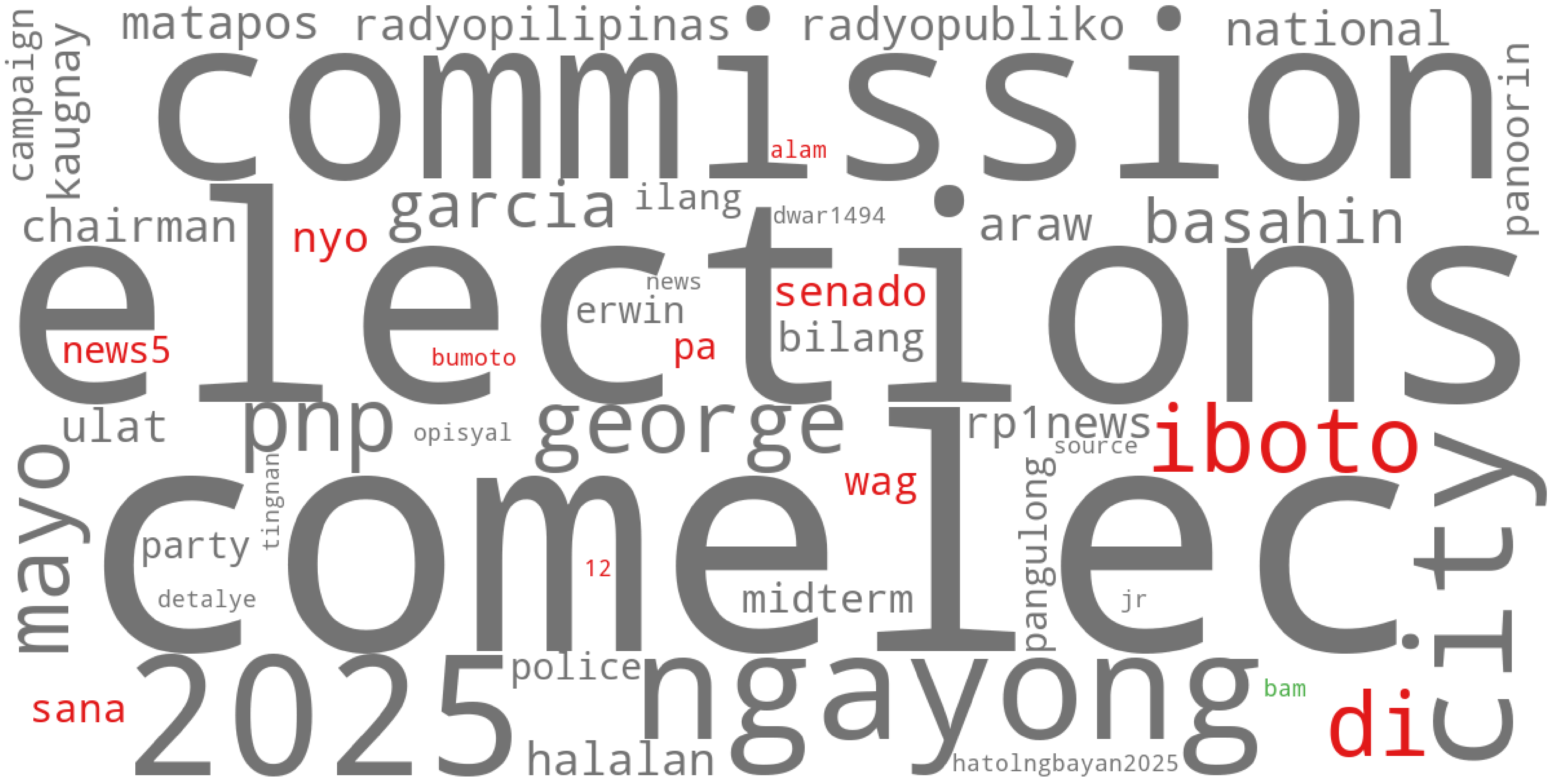
Results

Duterte Loyalist / Anti-Administration Rant Bloc



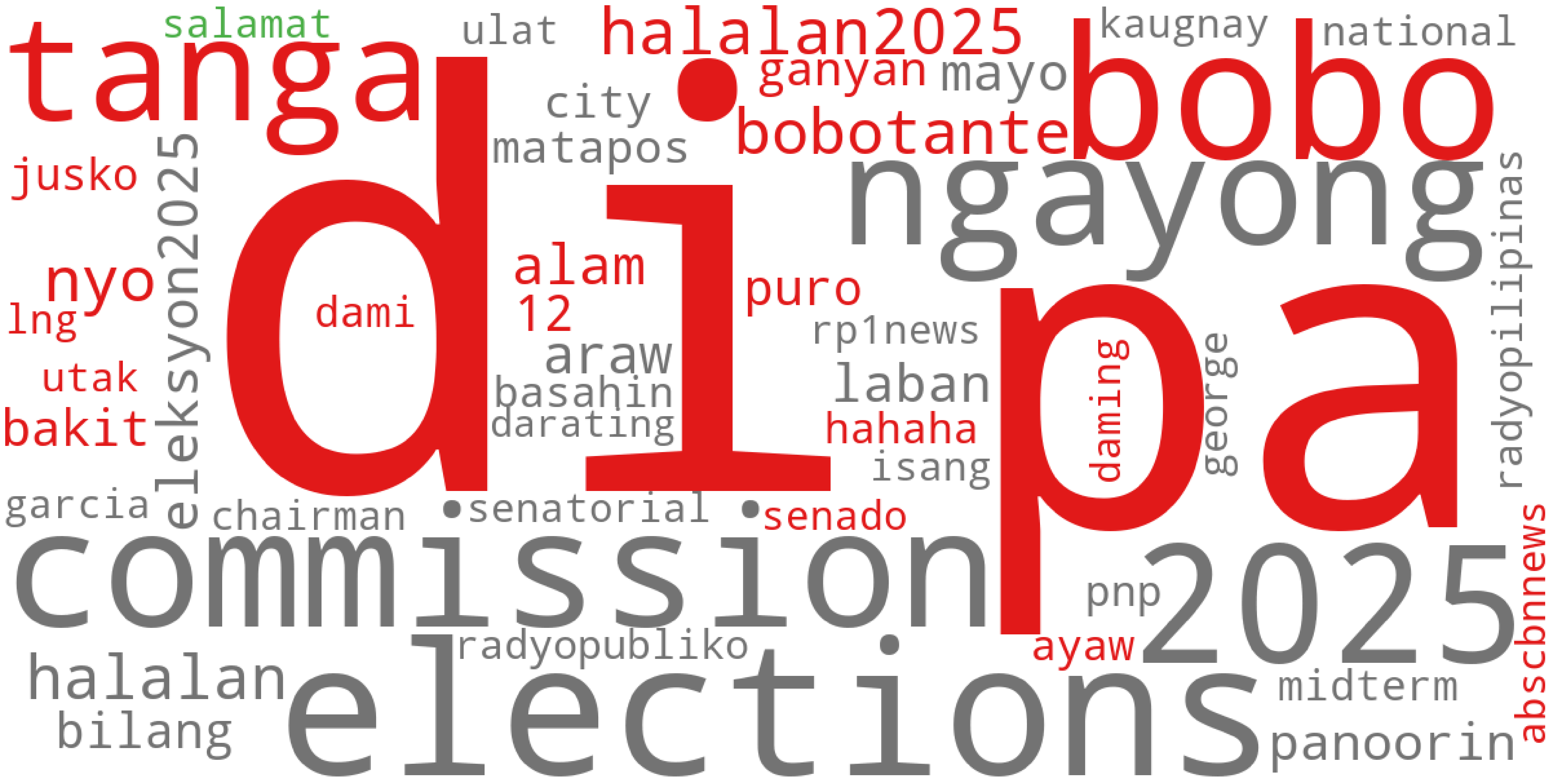
Results

COMELEC \& Media / Election Process News



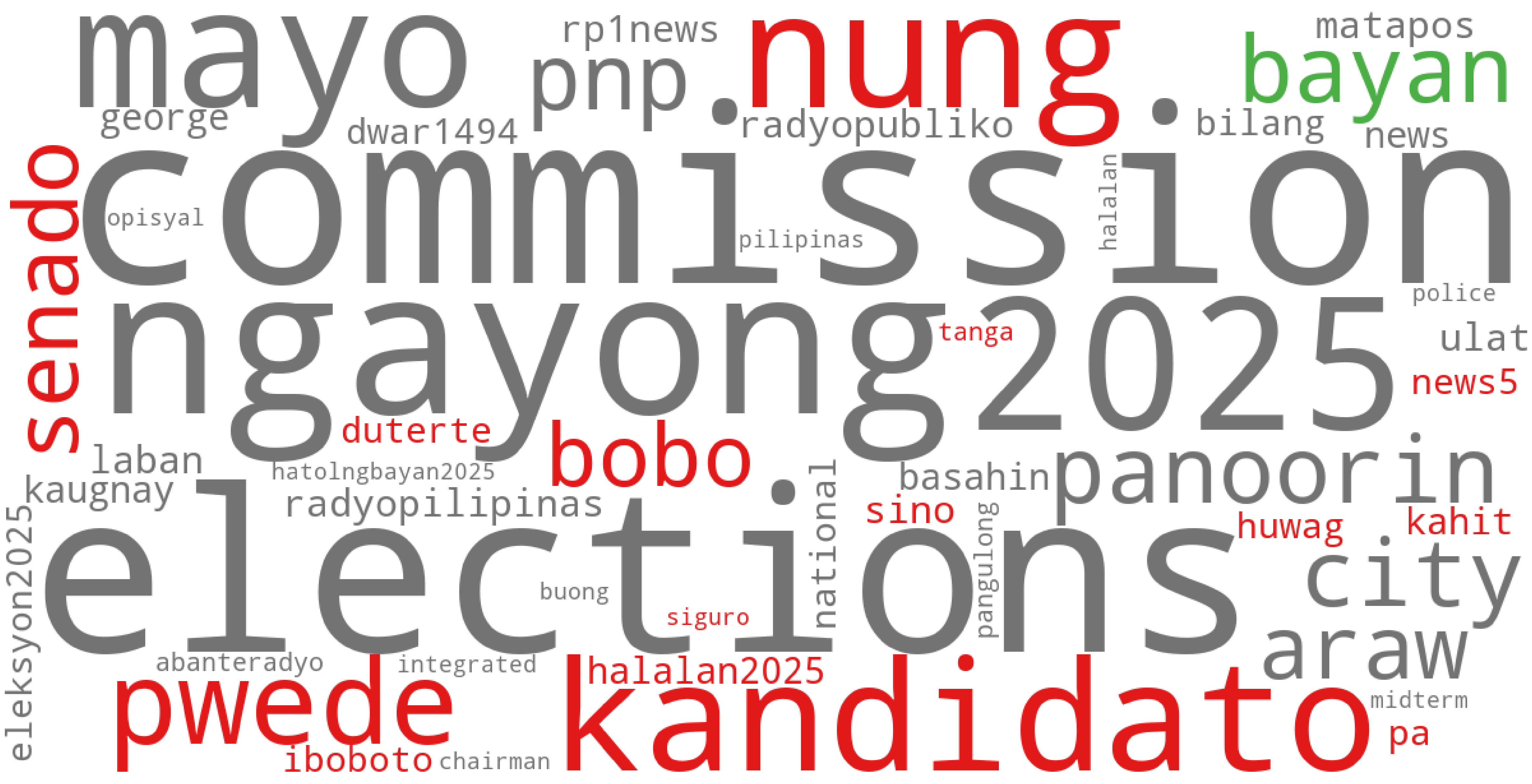
Results

"Bobotante" / Voter Frustration \& Mockery



Results

General Public / Undecided / Mixed Queries



Conclusion

NLP Contribution

Socio-Political Contribution

Future Work

Thank You

References

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